

Data Preparation, Cleaning and Classification for Price Statistics

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11:15-12:00 (Brazil Time)**

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on Scanner Data**

Agenda

- What is data preparation and why is it important?
 - Common data quality issues
 - Cleaning & standardization
- What is classification and how does it support CPI production?
 - Classification methods
 - Common challenges and considerations
- Data Filtering
- Practical Exercise
 - Dataset walkthrough
 - Tasks and Deliverables
- Questions



What is Data Preparation?

Transforming raw data into a structured, usable, and quality-assured dataset suitable for index production.

Transaction ID	Date	Product Description	Size	Price	Quantity	Department
1001	2026-01-05	COCA COLA 2L	2L	2.99	1	Beverage
1002	2026-01-05	Coke 2000ml	2000ml	3.09	1	Snacks
1003	2026-01-05	CKE CLAS2ltr		3.05	1	Beverages
1004	2026-01-05	Coca Cola Pk	4L	6.50	1	Drinks
1005	2026-01-05	Coca Cola 2L	2L	2.99	-1	Beverage
1005	2026-01-05	Coca Cola 2L	2L	2.99	-1	Beverage



Common Challenges in Administrative Data

VOLUME

Millions of records
per delivery

STRUCTURE

Semi-structured or
unstructured
product
descriptions

METADATA

Incomplete or
inconsistent
documentation

CONTINUITY

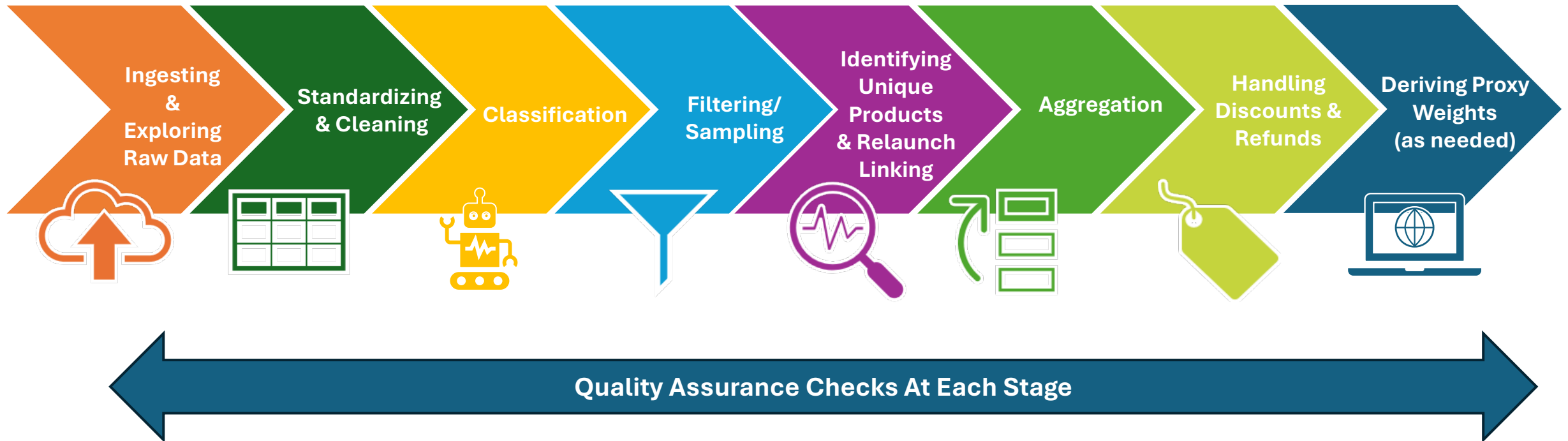
Changes in
content or
coverage over time

MEASUREMENT

Limited weighting
and quality
information



High Level Data Preparation Steps



Ingesting and Exploring Raw Data

The ingestion stage ensures that:

- Data have been delivered as expected.
- Data structure and content are internally consistent.
- Coverage and continuity are understood.
- Major anomalies are identified early.



Incomplete or delayed deliveries



Structural inconsistencies across files or time



Hidden truncation or encoding issues



Sudden coverage shifts



Undetected extreme values



Early detection
prevents
downstream
propagation and
costly rework



Cleaning & Standardization

Data Cleaning	Data Standardization
Focuses on improving data quality	Focuses on improving data consistency
Identifies and addresses errors, missing values, duplicates, and invalid observations	Converts data into a common format, structure, or representation
Helps ensure observations are valid and suitable for analysis	Helps ensure observations can be compared consistently
Example: Remove records with negative prices	Example: Convert "2000 ml" and "2L" to a common unit



Data Cleaning

Identifying and correcting issues that may affect data quality, analysis, or index calculation.

Issue	Example	Common Approaches
Missing values	Missing price or quantity	Remove records, impute values, or flag for review depending on importance and frequency
Invalid values	Negative price, quantity = 0, impossible dates	Apply validation rules and remove or correct invalid observations
Duplicate records	Same transaction appears multiple times	Deduplicate using transaction IDs, product IDs, timestamps, or matching rules
Outliers	Extremely high or low prices	Investigate, apply plausibility thresholds, compare with historical values, or flag for review
Out-of-scope products	Non-consumption goods, services outside CPI scope	Filter using classification codes, keywords, or product lists
Returns and test transactions	Scanner record showing negative sales	Exclude using retailer flags or business rules
Unavailable web products	Placeholder listing or out-of-stock item	Remove using availability indicators or scraping rules

Standardization

Product descriptions

Extract relevant characteristics and ensure descriptions can be consistently interpreted and processed

Units of Measure

Convert to standard (e.g., 1L, litre)

Price

Align currency formats, ensure prices are represented consistently

Date/Time

Convert to consistent format (e.g. YYYY-MM-DD)

Store/Outlet IDs

Ensure unique identifiers for outlets

GTINs/Product ID

Check for invalid or truncated codes



Standardization Tools and Techniques

Technique	Description	Common Tools
String Cleaning	Remove formatting inconsistencies in text fields	Regular expressions (regex), string manipulation libraries (e.g., stringr in R, string methods in Python), find-and-replace functions
Unit Conversion	Convert units to a common standard (e.g., 2000 ml, 2 litre, and 2L → 2 L)	Lookup tables, conversion dictionaries, regex parsing, unit conversion libraries
Date Parsing	Standardize date formats	Date/time libraries (e.g., lubridate in R, datetime in Python), standard date formats
Product Description Parsing & Normalization	Extract brand, size, flavour, package quantity, etc.	Regular expressions, keyword dictionaries, text processing libraries, natural language processing (NLP) techniques
Reference Tables & Lookup Files	Standardize values and map categories	Concordance tables, mapping files, reference databases, joins/merge operations
Variable Naming Standardization	Align datasets to a common schema (e.g., "price", "p", and "unit_price" → "price")	Data dictionaries, metadata standards, schema mapping tables

What is Classification



Classification is a mandatory prerequisite to be able to calculate elementary prices indices.



We need to assign prices in alternative data to the categories or the geographies they should belong to.



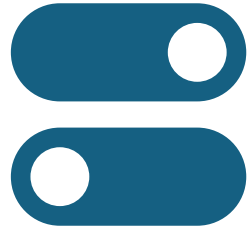
There are 2 types of classification necessary:

Product classification: unique products assigned to categories of the CPI (typically NSO specific category below COICOP5)

Geography Classification: Mapping outlets in the alternative data to the geographic strata of interest.



How to approach classification?



You typically need to do:

‘initial classification’ - i.e. classify historic data before you begin your research
operationalize **‘recurrent classification’** - i.e. to classify all new unique products each period

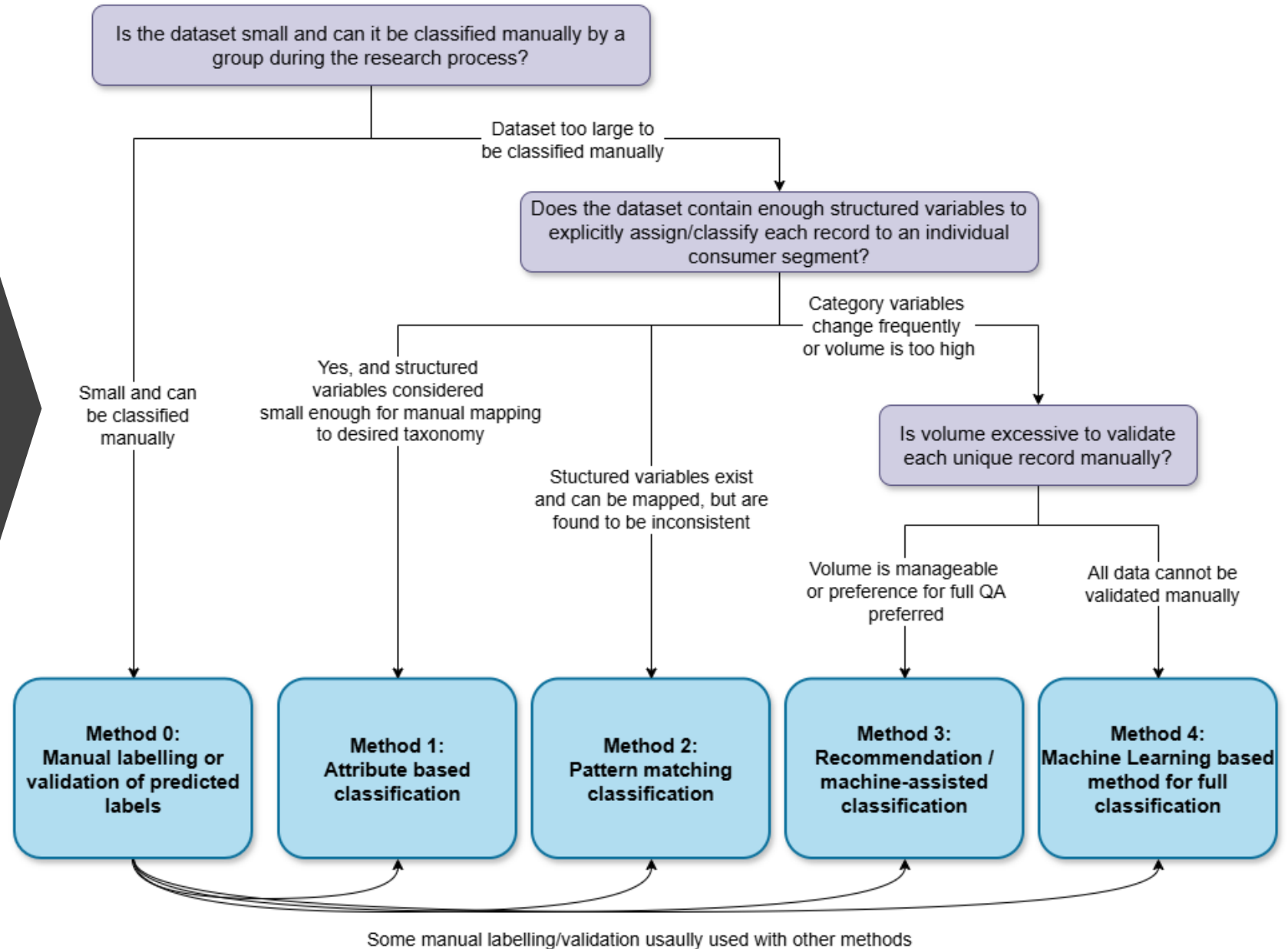


There are several classification methods depending on

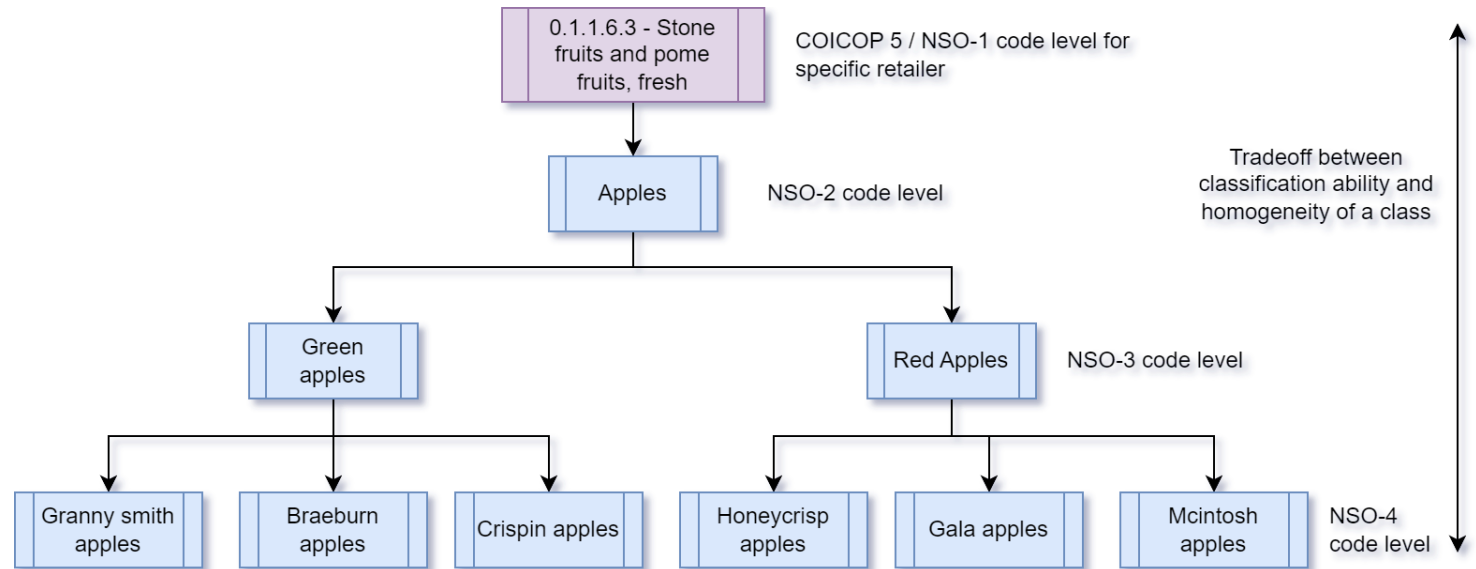
The scale of your initial or recurrent work
the complexity of the classification
quality of the data
how granular you want to classify to
etc



Decide on the appropriate classification method



Example of method 2 (pattern matching)



Consider you want to classify apples (retailer category) and product name using key words. Your code could be something like:

Simple example of attribute based classification

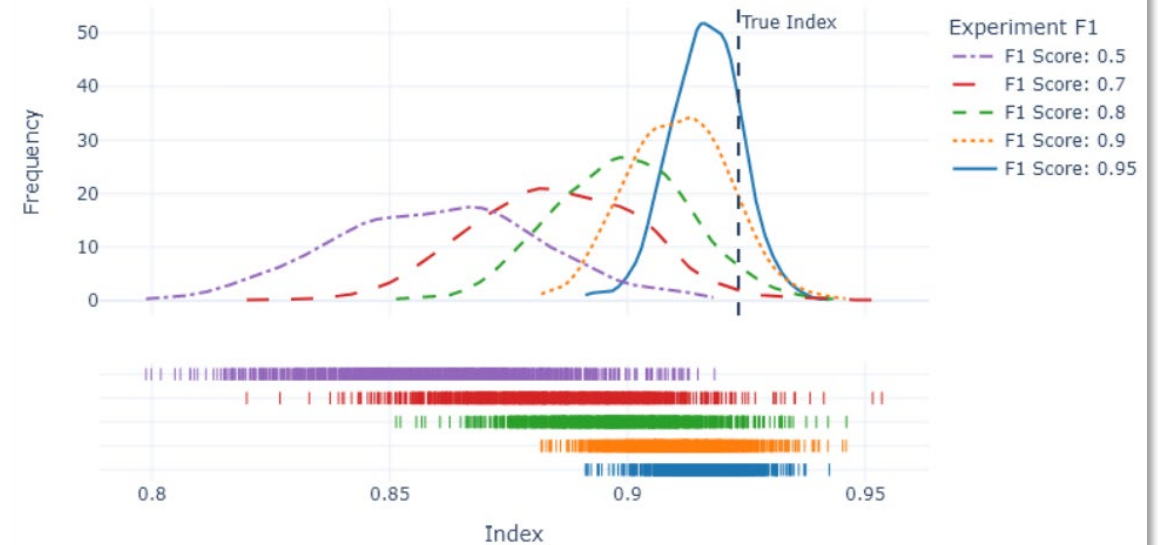
```
if alternative_data_category = 'Apples'
    if product_name includes 'GALA'
        coicop_code = 'red apples'      #or code for red apples category
    if product_name includes 'Crispin'
        coicop_code = 'green apples'    #or code for green apples
```

ML is an option, however it carries risks/costs!

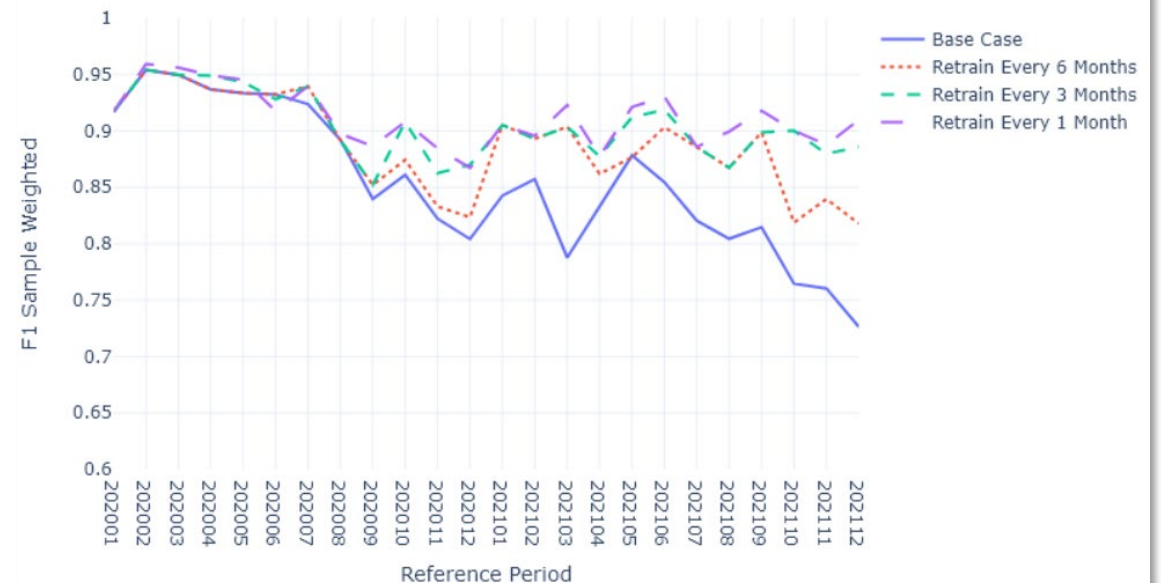
- ML models are never perfect – and misclassification can cause measurement error. Hence some human validation is always needed. See [Spackman et al \(2023\)](#) for some examples.
- When evaluating models, focusing on precision helps minimize misclassification bias, focusing on recall helps minimize variance. See [Spackman et al \(2024\)](#).
- Data drifts over time, hence model performance can degrade over time. Need to have retraining processes in place.
- To operationalize ML properly, need to invest in ML Operations (i.e. MLOps) processes. Check out [Ritter et al \(2024\)](#) for Statistics Canada's MLOps system.

Both from [Spackman et al \(2023\)](#)

Impact of Random Misclassifications on Calculated Price Relatives (Single EA)



Impact of Model Retraining on Model Performance Decay: Retailer 2



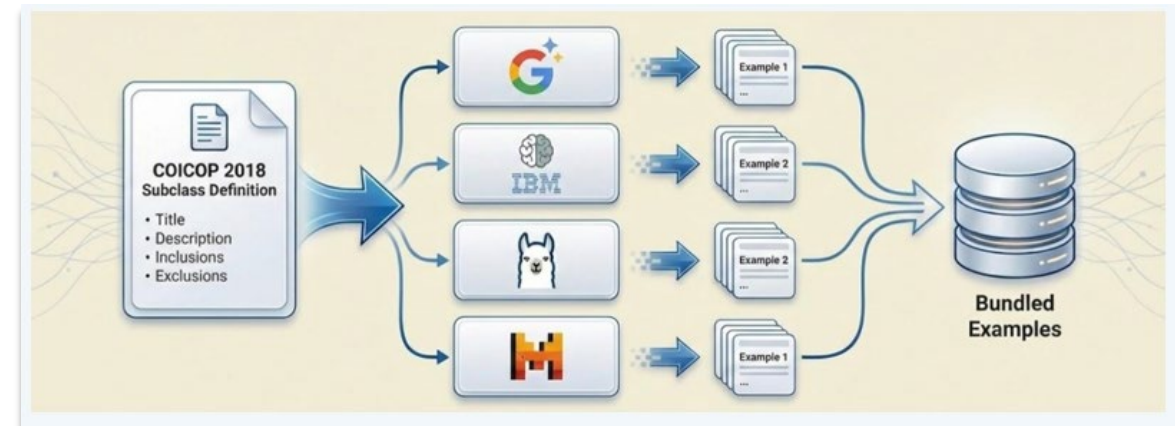
What about LLMs?

- Classification is a ‘supervised’ use case – i.e. there is a correct answer (even if annotators sometimes disagree).
- Large Language Models (i.e. LLMs) have been tried by a few and they can work
 - [Spackman et al \(2026\)](#) – used LLMs to classify each level of the CPI hierarchy (directly); and extracting structure data from unstructured data (e.g. extracting regular fields from web-scraped html files).
 - [Inês et al \(2026\)](#) – used Retrieval-Augmented Classification and got typical accuracies in the 60-80% range.
- Cost of prediction may not be trivial if scaled to large cases – LLM tokens aren’t cheap 😞.
- Possible uses of LLMs:
 - May be a robust way to make the initial annotation datasets. Validation/testing datasets should however be robustly annotated to have an accurate assessment of performance
 - Can support the annotation/validation process – be part of the recommendation processes for instance

Classification Precision by Method and Retailer



From [Spackman et al \(2026\)](#)



From [Inês et al \(2026\)](#)

Data Filtering

Data Filtering:

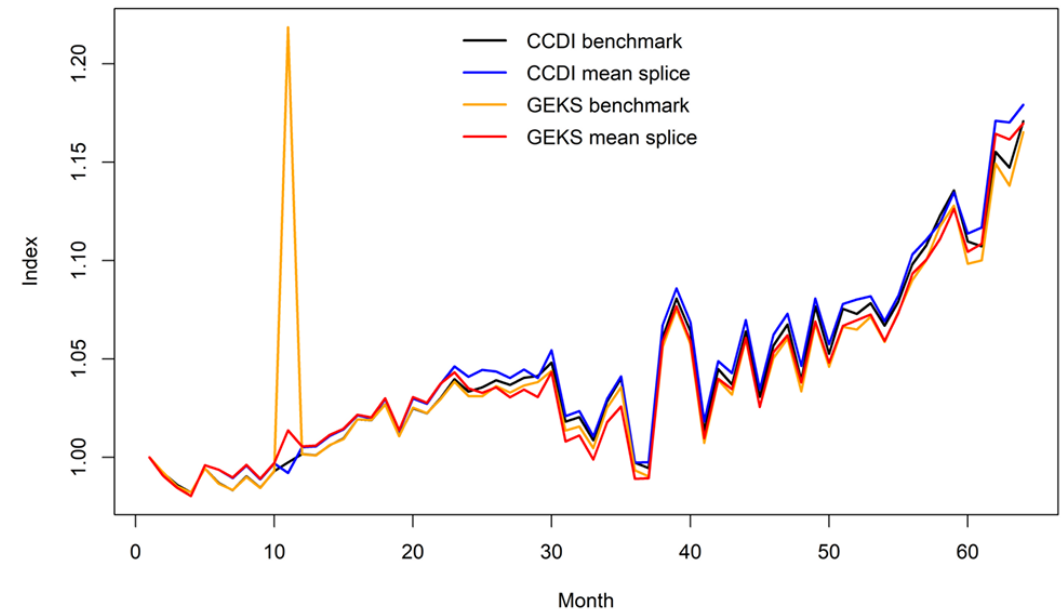
The process of applying predefined rules to identify observations that should be excluded, treated separately, or given less influence in index calculation

- Improves accuracy and reliability
- Reduces noise and improves clarity
- Reduces bias
- Enhances data integrity

Types of Data Filtering:

- Outlier filter: Aims at excluding or correcting extreme price increases or price decreases, typically compared to the previous period, from the price index calculation.
- Dumping filter: Aims at eliminating the downward pressure of clearance prices on the index.
- Low sales filter: Ensures that products with small expenditure shares do not unduly influence the index results.

CCDI and GEKS Indexes for Grooming Products



From [Fox \(2026\)](#)



Practical example

Checkout:

[Data preparation, cleaning, and classification – Brazil Workshop on the use of Alternative Data Sources for Consumer Price Statistics](#)



Practical Activity

Scenario: Building on the practical example, your task is to replicate and expand the exploration. You can use [Dominick's scanner data](#), [Polish open scanner data](#), [New Zealand electronics scanner data](#), or other data of your choosing.

Task

- Exploring the data to understand how to create data ready for aggregation (across time, outlets, and products)
- Clean obvious issues Standardizing appropriate columns
 - Idea: if you explore other Dominick's categories, take a shot at cleaning unit of measure column (SIZE column in the UPC files)
- Filtering appropriate values
- Classifying products
 - Ideas: If you want to expand on the rule based Dominick's example, you can (1) further refine how you find the most notable terms (hint, perhaps a sorted list of keywords per category will tell you which is most useful), (2) try to refine classification by using regular expressions, or (3) try a basic sklearn pipeline with a classifier.



Deliverable

Use Google Colab to create two outputs (i.e. deliverables):

1. Cleaned Dataset (i.e. what you process with your code).
2. Data Preparation documentation (include detailed overview to practice documenting as if you would use it for production)
 - What cleaning was handled and what was not:
 - What data was filtered out (which could be a data issue to keep an eye on from a data provider point of view)?
 - What data issues could escape the cleaning methods you designed (what to look out from a production point of view)?
 - What standardization rules were applied and how could they break in production?
 - What is the accuracy of the classifier and how sensitive do you think it is to data issues? What to look out for?



Questions?



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